

SARDAR PATEL UNIVERSITY
BCA (SEMESTER - II) EXAMINATION

Monday , 27th March, 2017
MATHEMATICS : US02FBCA02
(Mathematics)

Time : 10:00 a.m. to 12:00 noon

Maximum Marks : 70

Que.1 Fill in the blanks.

10

- (1) Edges connecting the same end points are called
(a) trivial graph (b) multigraph (c) loops (d) multiple edges
- (2) The sum of the degrees of the vertices of a graph is the number of edges.
(a) greater than (b) less than (c) equal to (d) equal to twice
- (3) The diameter of complete graph K_n is
(a) $n - 1$ (b) 1 (c) $2n$ (d) n
- (4) In a map, the total degrees of the regions =
(a) $3e$ (b) e (c) $2e + e$ (d) $2e$
- (5) The chromatic number of graph K_{120} is =
(a) 60 (b) 30 (c) 20 (d) 120
- (6) In a connected map with $R = 11$, $E = 27$ then $V =$
(a) 20 (b) 19 (c) 36 (d) 18
- (7) $\binom{n}{n-7} =$
(a) $\binom{n}{n+7}$ (b) $\binom{n}{n}$ (c) $\binom{n}{7}$ (d) $\binom{n}{0}$
- (8) We can select four objects from the given 9 objects in ways.
(a) $\binom{4}{9}$ (b) $\binom{9}{4}$ (c) $\frac{9!}{4!}$ (d) $\frac{9!}{5!}$
- (9) The quartile Q_2 is coincides with
(a) Mean (b) Mode (c) Median (d) Standard deviation
- (10) Bowleys coefficient of skewness is defined as
(a) $\frac{Q_3 + Q_1 - 3Q_2}{Q_2 - Q_1}$ (b) $\frac{Q_3 - Q_1 + 3Q_2}{Q_3 - Q_1}$ (c) $\frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$ (d) None of these

Que.2 Answer the following (Any Ten)

20

- (1) Draw a graph $G_2 = (V_2, E_2)$, where $V_2 = \{P, Q, R, S, T\}$ and $E_2 = \{PQ, PR, PS, PT, TR, PR\}$. Is it simple?
- (2) For sequence of vertices $abcefdab$, state whether or not it represents a trail, path, closed walk or cycle in the graph illustrated.
- (3) Define the terms simple path and full subgraph with example.
- (4) Define spanning tree and give an example of graph with three spanning trees.
- (5) Find the E using Eulers formula for the connected planar multigraph having $V = 5, R = 3$
- (6) State Welch-Powell algorithm for painting a graph and give one example of it.
- (7) Write $\frac{n(n-1)(n-2)(n-3)\dots(n-r+1)}{1.2.3.4\dots r}$ in terms of factorial
- (8) Find n , if $P(n, 4) = 42 P(n, 2)$.

(9) Find the number of five letter words which contain three different consonants and two different vowels.

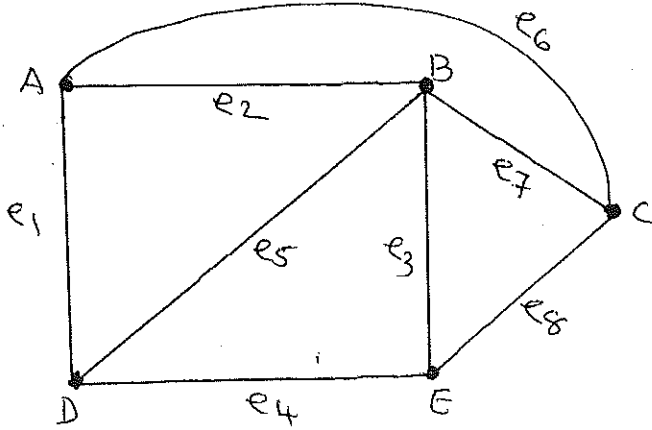
(10) Simplify $\frac{(n-r+1)!}{(n-r-1)!}$.

(11) Define Karl Pearsons coefficient of correlation and state its properties.

(12) Explain the linear regression and state its properties.

Que.3 (a) Find the incidence matrix and adjacency matrix for the following graph .

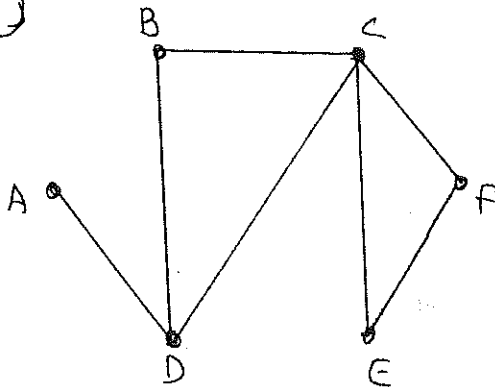
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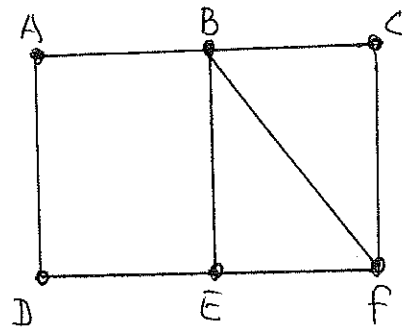
(b) Define bridge and cut points. Find bridge and cut points for the following graph.

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(i)



(ii)



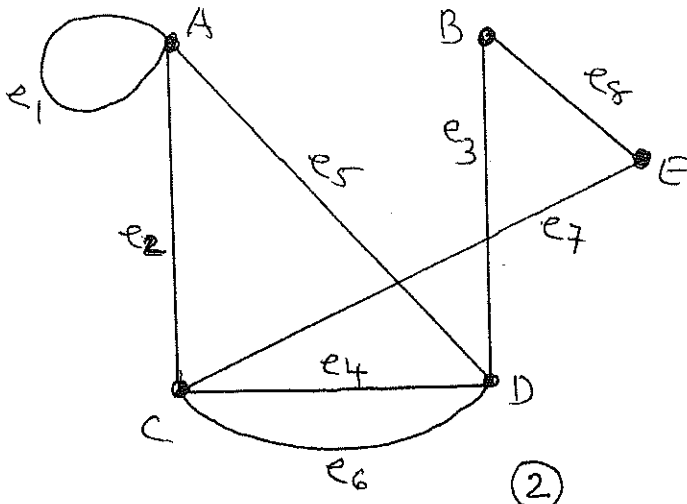
OR

Que.3 (a) Define complete graph. Draw complete graph with seven vertices and eight vertices.

5

(b) Find the incidence matrix and adjacency matrix for the following graph.

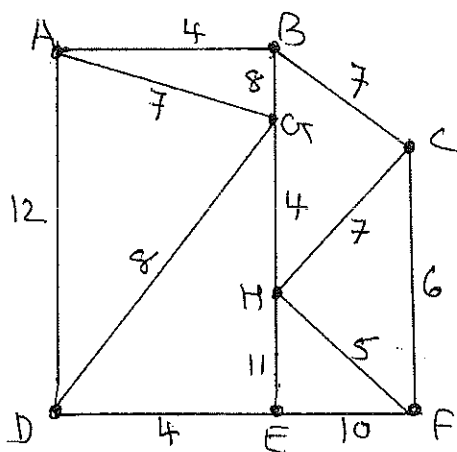
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(2)

Que.4 (a) Find the number of spanning trees in each of the following graph. Also find minimal spanning tree.

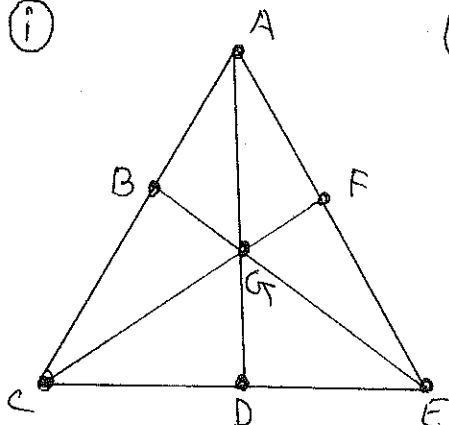
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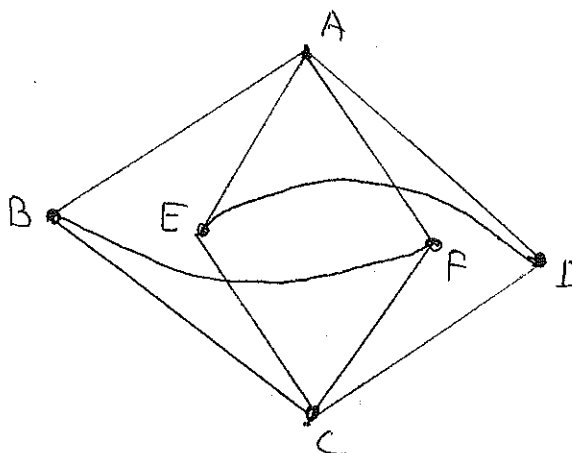
(b) What do you mean by the chromatic number of the graph? Determine the chromatic number of each of the following graphs.

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(i)



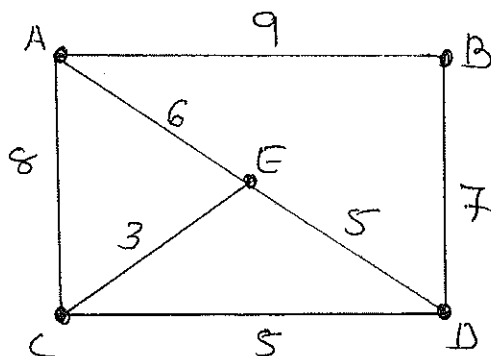
(ii)



OR

Que.4 (a) Find the number of spanning trees in each of the following graph. Also find minimal spanning tree.

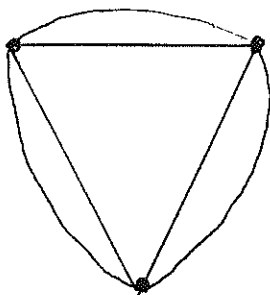
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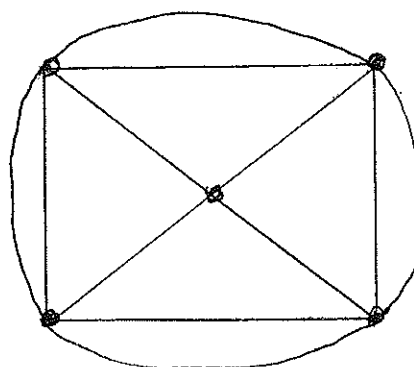
(b) Verify Eulers formula for the following graphs.

5

(i)



(ii)



(3)

(P.T.O.)

Que.5 (a) Find n if $2P(n, 2) + 50 = P(2n, 2)$.

(b) Find the number of ways that four mathematics books, three history books, three chemistry books and two sociology books can be arranged on a shelf so that all books of the same subject are together.

(c) Suppose repetitions are not permitted. (i) Find the number of three-digit numbers that can be formed from the six digits 2, 3, 5, 6, 7, and 9. (ii) How many of them are less than 400? (iii) How many of them are even?

OR

Que.5 (a) Find the number m of Permutations that can be formed all the letters of the word MISSISSIPPI.

(i) Solve it if the word are to begin with P.

(ii) Solve it if the word are to begin and end with I.

(b) A debating team consists of 3 boys and 3 girls. Find the number of ways they can sit in a row where: (i) there are no restrictions (ii) the boys and girls are each to sit together (iii) just the girls are to sit together.

Que.6 (a) Given below is the frequency distribution of the marks obtained by 90 students. Compute the standard deviation.

Marks	20-29	30-39	40-49	50-59	60-69	70-79	80-89	90-99
No. of students	5	12	15	20	18	10	6	4

(b) Calculate Karl Pearsons coefficient of correlation between x and y from the following data:

x	10	6	9	10	12	13	11	9
f	9	4	6	9	11	13	8	4

OR

Que.6 (a) In an experiment the number of grams of a given salt which dissolved in 100g of water was observed at eight different temperatures. The weights of salt at different temperatures are given below.

Temp(degree C)	0	10	20	30	40	50	60
Weight of salt(g)	51.1	61.5	67.2	72.6	73.5	82.2	83.2

Find the regression equation which could be used to predict the weight of salt given the temperature. Predict the weight of salt which would dissolve at temperatures (i) 25°C (ii) 85°C

(b) Frequency distribution of the blood pressure given below, Compute the quartile Deviation of the frequency distribution.

Systolic BP (mm Hg)	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of Infants	1	6	14	43	21	13	10	1



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(4)